

José Arbelaez

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EDUCATION

- Universidad Autónoma de Madrid | BSc in Data Science and Engineering** 2022 – 2026
- Received highest honors in Natural Language Processing, Computer Vision, Computer Networks, High-Performance Computing, Statistical Methods and Machine Learning.
 - Board Member of the university Computer Security Club (SEIF UAM), 2023 – 2024.
- Google Cloud | Cloud & Data Engineering Certificate** 2025 – Present

EXPERIENCE

- Naudit HPCN | Data Scientist Intern** Jan 2026 – May 2026
- Managed Logstash, Filebeat and Elastic Agent pipelines processing more than 50 TB of banking-platform data per day, correcting production errors and monitoring the system with Grafana and Prometheus.
 - Developed an alarm autocorrelation system using 17 statistical features, calibrated HistGradientBoosting models, SHAP explanations and causal DAG visualization, achieving F1 above 0.80.
- Google Developer Group on Campus UAM | President & Co-Founder** Jul 2025 – Present
- Co-founded the chapter, coordinate a multidisciplinary team and lead AI, cloud and software-development events for more than 200 students; designed gdguam.es and helped implement several of its components.
- VULCASAN SL | Full-Stack Developer** Jun 2024 – Jul 2025
- Built a production application automating client, inventory and operational workflows for an automotive workshop network; developed backend APIs, authentication and CI/CD in a four-person team.

SELECTED PROJECTS

- Bachelor's Thesis | Surrogate Models** [\[Read article\]](#) 2025 – 2026
- Built a Gaussian-process and Expected-Improvement framework for UAM's Healthy Food Ingredients Group to prioritize untested *Hermetia illucens* diets from scarce data, reducing the search space for costly food-biology experiments.
- RoadGuard | Reliable Road-Damage Detection** [\[Read article\]](#) 2026
- Turned crack and pothole detection into a deployment-oriented inspection system with cross-country validation, calibrated confidence, abstention and explainable maintenance prioritization.
- BGG Review Intelligence | NLP & Opinion Mining** [\[Read article\]](#) 2025
- Built a reproducible NLP pipeline that converts unstructured BoardGameGeek reviews into sentiment predictions and aspect-level evidence for product and player-experience decisions.
- Image Generation with Diffusion Models** [\[Read article\]](#) 2025
- Implemented VP, VE and Sub-VP diffusion models from scratch on CIFAR-10, covering score matching, multiple samplers, conditional generation and image imputation.

HONORS & AWARDS

- 1st of 240+ participants – UAM Investors League**; achieved the competition's highest Sharpe ratio with a risk-adjusted portfolio strategy, UAM and IronIA Fintech (2025).
- 2nd Place – Cloud Agent Systems Hackathon**, Google Cloud and Diverger (2025).
- Academic Excellence Scholarship**, Community of Madrid (2024 and 2025).

LANGUAGES & TECHNICAL SKILLS

- Languages:** Spanish – Native; English – C1; French – Basic.
- AI/ML:** Scikit-learn, PyTorch, TensorFlow/Keras, Gaussian processes, Bayesian optimization, SHAP, computer vision and NLP.
- Data & Infrastructure:** PostgreSQL, DuckDB, MongoDB, Elasticsearch, Logstash, Grafana, Prometheus, Docker, Linux, Git and GCP.

ADDITIONAL INFORMATION

- Interests & Activities:** Open-source contributions, building practical hardware and automation for my home garden, and playing the harmonica.